

Bidialectal and monodialectal differences in morphosyntactic processing of AAE and MAE: Evidence from ERPs and acceptability judgments

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ARTICLE INFO

Keywords:

EEG
Black English
ERP
P600
AAE
African American English
Dialect
Language
Grammar
Morphology
Syntax
Bidialectalism
Bilingualism

ABSTRACT

Introduction: African American English (AAE) has never been examined through neurophysiological methods in investigations of dual-language variety processing. This study examines whether contrastive and non-contrastive morphosyntactic features in sentences with and without AAE constructions elicit differing neural and/or behavioral responses in bidialectal speakers of AAE and Mainstream American English (MAE), compared to monodialectal MAE speakers. We compared electroencephalographic (EEG) and behavioral (grammatical acceptability judgment) data to determine whether two dialects are processed similarly to distinct languages, as seen in studies of bilingual codeswitching where the P600 event related potential (ERP) has been elicited when processing a switch between language varieties.

Methods: Bidialectal AAE-MAE speakers ($n = 15$) and monodialectal MAE speakers ($n = 12$) listened to sentences in four conditions, while EEG was recorded to evaluate time-locked brain responses to grammatical differences between sentence types. The *maintained verb* form in the present progressive tense sentences (e.g., *The black cat lap/s the milk*) was the morphosyntactic feature of interest for comparing P600 responses as an indicator of error detection. Following each trial, responses and reaction times to a grammatical acceptability judgment task were collected and compared.

Results: Findings indicate distinct neurophysiological profiles between bidialectal and monodialectal speakers. Monodialectal speakers demonstrated a P600 response within 500–800ms following presentation of an AAE morphosyntax feature, indicating error detection; this response was not seen in the bidialectal group. Control sentences with non-contrasting grammar revealed no differences in ERP responses between groups. Behaviorally, bidialectal speakers showed greater acceptance of known dialectal variation and error (non-contrastive) sentence types compared to the monodialectal group.

Conclusions: ERP and behavioral responses are presented as preliminary evidence of dual-language representation in bidialectal speakers. Increased consideration of AAE language

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processing would enhance equity in the study of language at large, improving the work of clinicians, researchers, educators and policymakers alike.

1. Introduction

Mainstream American English (MAE) (also known as General American English or Standard English) is the variety of English used by the dominant culture in the United States and is the educational standard for reading and writing instruction. African American English (AAE) is the most widely used dialectal language variety of American English (AE), and is spoken by or among African Americans (AA) and various urban and Southern American English communities, with written forms often used to reflect the experiences of its speakers in literary and musical works (Schneider et al., 2004; Wolfram & Thomas, 2008; Young et al., 2018).

The study of AAE has shifted the collective discussion around non-mainstream American English (NMAE) dialects. As a whole – and not without conflict – the fields of speech-language pathology and education have formally moved from a deficit-based position on minoritized language varieties to a sociolinguistic one that purports to uphold their equal value to mainstream varieties, leading to changes in training for clinicians and educators (ASHA Committee on the Status of Racial Minorities, 1983; American Speech-Language-Hearing Association [ASHA] 2003, 2020; Congressional Research Service, 1993; Seymour et al., 1999).

Still, a paradox exists at the heart of research on bilingualism and bidialectalism: while many dialects are understood as separate, rule-governed systems, these language varieties are often not the focus of the same questions around dual-language use as the *standard* language varieties. Unless the aim is the examination of dialect, language studies with typical and/or disordered populations do not regularly mention dialect as a factor in participant selection or language background. Neurophysiological investigations that include dialects in their discussion of language processing are especially rare. Despite earnest efforts to dismantle oppressive systems faced by minority dialect speakers in the US, special service provision, education, and research endeavors will remain stalled in a state of comparison to the performance of MAE until bidialectal speakers are studied in the same manner as bilingual speakers (Hamilton et al., 2018; Otting et al., 2016; Reaser et al., 2017; Young, 2020).

Most studies of dialect have been conducted using behavioral assessments measuring production and comprehension of contrastive and non-contrastive features, in order to inform educators, speech language pathologists (SLPs), and researchers how to identify profiles of typical acquisition and potential delay/disorder between and within dialects. No studies to date have investigated the neurophysiological nature of bidialectal sentence processing in speakers of AAE and MAE.

Using electroencephalography (EEG), the current study examines how contrastive and non-contrastive morphosyntax features between AAE and MAE are processed in speakers of AAE and MAE through the recording of neurophysiological responses to different morphosyntax features in AE, alongside behavioral responses (acceptability judgment tasks and response times). In addition to providing an example of how AE dialect research could be furthered in the SLP and neuroscientific literature, this work hopes to contribute to the budding discussion of neuroscientific theories of bilingualism, as it relates to dialects (Grimaldi, 2018; Zaharchuk et al., 2021).

1.1. EEG in AAE research

EEG was chosen for this study to provide insights into language processing, because it offers a data collection method not reliant on behavioral responses or on conscious awareness of dialectal features and use. This is important with AAE speakers because they may not identify with a language variety treated as *low prestige* (Ferguson, 1959), whereby persistent discrimination is associated with its use, or even with just being perceived as a speaker (for some examples, see DeMeis & Turner, 1978; Massey & Lundy, 2001; Rickford & King, 2016). Alternatively, while many speakers of AAE proudly identify with use of its linguistic features, they may not align themselves with the term *African American English* (Mufwene, 2001). By using a time-locked listening paradigm during EEG recordings, event-related potential (ERP) signatures can be identified as participants listen to sentences, regardless of their consciously-held beliefs or decisions about the grammatical structure of those sentences.

1.2. The P600 in studies of dual-language variety processing

The P600 is the ERP component targeted in this study, also referred to in other work as the late positive component (LPC) or syntactic positive shift (SPS). In language research, the P600 has been identified as a response to ungrammatical, anomalous and dispreferred sentence constructions, and is thought to represent brain activity related to repair and reanalysis (e.g., Kaan et al., 2000; Leckey & Federmeier, 2020).

Support for the P600 as a target here was derived from investigations that associated distinct ERP signatures with codeswitching between traditionally distinct language varieties (e.g., Spanish and English), as well as between two language varieties generally considered to be dialects of the same language (e.g., diglossia in Arabic). Moreno et al. (2002) compared between-language shifts of English-Spanish bilinguals to within-language lexical switches in response to written text stimuli. Their findings indicated a similarity of processing between codeswitching and morphosyntactic operations, suggesting that the P600 response to codeswitched conditions may be associated with a shift between language varieties.

Similar to AAE and MAE, diglossic speakers of formal and colloquial Arabic varieties do not always identify as speaking different dialects, despite different performance in each language variety (Eviatar & Ibrahim, 2000; Khamis-Dakwar, 2020). Cross-linguistic

switches between dialect varieties within a sentence were found to significantly enhance P600 effects, similar to the English/Spanish between-language switch response demonstrated previously by Moreno et al. (2002, 2008). Khamis-Dakwar and Froud's (2007) findings supported a view of two varieties of Arabic as representing two separate language systems in the brain, despite sociocultural beliefs that they exist as different registers along a continuum of the same language.

To date, there have been few ERP studies investigating NMAE dialects. Conrey et al. (2005) examined phonological processing of a non-peripheral front vowel merger before a nasal consonant, present in Southern American English (SAE) dialect speakers. Bidialectal speakers of this merged vowel dialect would be perceived by monodialectal MAE speakers as producing the words *pɪn*/*pɛn* and *main*/*mine* as homonyms. In this experiment, simultaneous auditory and word-by-word text stimuli were presented in sentences to merged and unmerged dialect speakers (e.g., "Sign the check with a /pɪn/"). Results demonstrated elicitation of an LPC/P600 in unmerged dialect speakers following the onset of the incongruent target word. Merged dialect speakers also demonstrated reduced behavioral discrimination effects (along with the LPC/P600) in the condition presenting contrasting vowels in words where a critical vowel merger might have been (e.g., when participants read the word *pin* but heard the word /*pen*/).

At present, there are no ERP investigations of AAE bidialectal speakers specifically, and few relating to syntax in speakers who might use dialect-specific features also included in AAE. Two recent investigations have examined how familiarity with a dialect can modulate online responses and/or later decision-making about sentences with and without dialect-specific features (Weissler & Brennan, 2020; Zaharchuk et al., 2021). Weissler and Brennan (2020) presented auditory sentences that were produced by a single bidialectal speaker, with phonology and prosody in either AAE or MAE (referred to as *African American Language* [AAL] and *Mainstream U.S. English* [MUSE], respectively). Recorded words were spliced together to create the described conditions within each dialect presentation. Sentences were presented in one of three conditions: (1) where morphosyntax features of AAE and MAE were non-contrastive and acceptable to all speakers; (2) where morphosyntax was contrastive and acceptable to AAE speakers only; and (3) where morphosyntax features were non-contrastive and ungrammatical to all speakers. Interestingly, while participants did not demonstrate P600 responses to the first and second condition in the MAE-accented sentences, no P600 at all was demonstrated across conditions in the AAE-accented sentences. Weissler and Brennan therefore found mixed support for the hypothesis that speakers with familiarity of a dialect pre-consciously alter their expectations of grammaticality depending on the speakers' language. Participant listeners in this study were mixed in terms of AE dialect background, however, which suggests that dialect-specific expectations may be different depending on the listening group. Or it may be that the stimuli still allowed for the perception of dialect features in the MAE-accented condition. However, this would not account for the lack of error detection in the overlapping ungrammaticality / AAE-accented condition, unless the expectation of *more* accented dialectal speech is that all manner of differences in morphosyntax increase in likelihood, including errors. Similar findings suggesting changing grammatical expectations during the processing of accented speech were reported by Hanulíková et al. (2012).

Zaharchuk et al. (2021) manipulated a Southern US English (SUSE) construction known as the double modal (e.g., *might could borrow*). Examples were auditorily presented in sentences to two groups of participants, those familiar and unfamiliar with this and other SUSE features. Findings demonstrated P600 elicitation in both groups for this construction compared to single modal sentences (e.g., *might borrow*), with behavioral judgments showing greater acceptability of double modals (even sensitivity to some types of double modals over other, less likely types) in the familiar group. Zaharchuk et al. interpreted this evidence for online language processing of dialect as indicating a preference for more frequent, higher prestige, mainstream language constructions over variants, with behavioral responses reflecting the impact of sociolinguistic exposure (Sumner et al., 2014). They suggested that, with increased familiarity to a variant structure, a frequency threshold might be crossed which then could be reflected in both online processing and behavioral acceptability of dialectal variation, thereby supporting more usage-based theories to language processing (Bybee, 2006; MacDonald, 2013).

The current study fulfills the need to specifically investigate the syntactic processing of a more linguistically-controlled group of bidialectal AAE-MAE speakers and monodialectal MAE speakers, using a contrastive feature that is considered a stable, high-frequency and widely-used dialect-specific feature as the point of comparison.

1.3. African American English morphosyntax and verb maintenance

Terms related to AAE features will be presented here with the intention of defining morphosyntactic characteristics as their own constructions, not as a comparative breaking away from MAE. To this end, the morphosyntactic feature of interest in this study is the *maintained verb* in the present tense in AAE (e.g., *The black cat lap the milk*). This feature has also historically been described as a deletion, omission or absence of the -s marker on the present tense verb, but these descriptions uphold the rhetoric of a deficit framework for dialectal language use and diminish the accuracy of AAE feature descriptions by defining them by their absence of MAE features (Dandy, 1991; Easton & Verdon, 2021; Hamilton et al., 2018; Stockman, 2010).

AAE speakers have been shown to maintain the uninflected form of the 3rd person present tense singular verb even when speaking to listeners outside of their community, making it a particularly robust and consistent feature of AAE usage across regional populations (De Villiers & Johnson, 2007). It is also a contrastive feature to MAE, in that the -s marker on a 3rd person singular verb is obligatory for typical, monodialectal MAE speakers. Description of this dialect-specific feature as a maintained verb is derived and adapted from Dandy (1991), as cited by Hamilton et al. (2018), with its function described mainly as one of retention/maintenance ("Third-person singular present form retains the same form in person and number," p. 114, Table 2, Hamilton et al., 2018).

Studies of language acquisition in typically-developing (TD) AAE-speaking children have demonstrated that constructions of verb maintenance are typical in bidialectal language acquisition (Craig & Washington, 1994; 2004; De Villiers & Johnson, 2007; Johnson, 2005). In a study comparing the verbal -s productions of typically developing (TD) and specifically language impaired (SLI) children,

Table 1

Conditions, group design, sample trials and predicted responses

Condition & sample trial		Monodialectal speakers' predicted responses	Bidialectal speakers' predicted responses
Block 1	1. Third person -s marked <i>The black cat laps the milk.</i>	Acceptable	Acceptable
	2. Third person maintained verb <i>The black cat lap the milk.</i>	Unacceptable (P600)	Acceptable* (MAE-aligned judgments and longer RT predicted)
Block 2	3. Accusative object <i>The gentle doctor comforts <u>them</u> in the clinic.</i>	Acceptable	Acceptable
	4. Nominative object <i>The gentle doctor comforts <u>they</u> in the clinic.</i>	Unacceptable (P600)	Unacceptable (P600)

Table 1. Conditions and example sentences, showing the acceptability of the morphosyntax feature in the given dialect. In the “unacceptable” contexts, the P600 component response is predicted. *Asterisk indicates a situation where, due to the academic setting, the bidialectal group was expected to provide MAE-biased behavioral responses to the heard sentences (i.e., judging them as unacceptable) despite this being an acceptable feature in AAE. In [Block 2](#), non-contrastive constructions between dialects were expected to elicit similar responses from both groups.

[Cleveland and Oetting \(2013\)](#) found that TD AAE-speaking children maintained the uninflected forms of nouns and verbs at a rate (22%) that was consistent with adult AAE utterances ([Labov & Harris \(1986\)](#) reported 10% to 50%, and [Wolfram \(1969\)](#) reported 29% maintenance). Such findings confirm that the maintained verb forms are a stable and prevalent morphosyntactic construction for most AAE-speaking populations, across regions, ages and generations.

1.4. Research questions and predictions

Investigations of the neural mechanisms underlying bidialectal language processing can connect researchers in education, speech-language pathology, and neurolinguistics, all looking for answers about people who speak languages unrecognized by the systems in which they live ([Oetting et al., 2019](#); [Young et al., 2018](#); [Zaharchuk et al., 2021](#)). The subsystems hypothesis of language representation holds that each language variety exists as a subset of neural connections, supported by a single cognitive system (i.e., the language system). Under this hypothesis, mixing and switching patterns for bidialectal speakers, who were raised in a context where both varieties are conceptualized as a single linguistic system, will demonstrate lower ‘interlanguage inhibition’ when processing sentences with contrasting features ([MacWhinney, 1997](#); [Paradis, 2004](#); [Wulfeck et al., 1986](#)). Hence, we predicted that bidialectal AAE-MAE speakers would demonstrate a P600 ERP response indicating error detection to sentences containing a shared grammatical violation of MAE, but not sentences that violate MAE grammatical constraints while being acceptable in AAE.

With respect to acceptability judgments, the dialect-shifting hypothesis is derived from evidence showing that bidialectal individuals will dialect shift in relevant situations (such as academic settings) where standard MAE is expected ([Charity et al., 2004](#); [Connor & Craig, 2006](#); [Craig & Washington, 2004](#); [Kohler et al., 2007](#); [Terry, 2006](#)). Therefore, AAE-MAE speakers were predicted to provide behavioral responses aligned with the responses of monodialectal MAE speakers to the acceptability judgment task.

2. Methods

2.1. Participants

Volunteers without neurological, hearing differences or language-related disorder were recruited from the New York City, NY area, and included students and local community members. Fifteen African American, bidialectal AAE-MAE speakers (9 Female, 1 Non-binary/Trans, 5 Male, mean age 29.8 years, age range 22-47 years) and twelve white American, monodialectal MAE speakers (7 Female, 0 Non-binary/Trans, 5 Male, mean age 28.6 years, age range 22-43 years) with no reported sustained exposure in childhood to other dialects or languages were eligible for participation in this study. Data from 3 participants (1 bidialectal and 2 monodialectal) were removed from EEG analyses due to noise artifact that resulted in fewer than 50% usable trials, but were included in the behavioral analyses, where task outcomes were unaffected.

Prior to the experiment, participants completed a language history questionnaire (see [Appendix B](#)), aimed at classifying dialect use and exposure during childhood (see [Mitchell, 2019](#); [Moody, 2011](#); [Parker & Reeves-Pelt, 2017](#) for use of self-reported cultural and linguistic background to determine eligibility of AAE dialect speakers). In order to limit variation and influence from other language varieties, participants who self-reported long-term exposure to a dialect other than AAE or MAE (e.g., Appalachian English, Jamaican Patois, Spanish-Influenced-English, etc.) as well as those who heard another standard language in their household from birth (e.g., German, Hebrew, French, etc.) were not eligible for the study. Report or evidence of spoken dialect was not required, since bidialectal speakers may not identify as speaking a dialect and/or may codeswitch in formal educational settings. Sources estimate that between 60 and 99% of African Americans use at least some features of AAE in their spoken communication ([Dillard, 1972](#); [Smitherman, 1977](#); [Spears, 1987](#); [Ulatowska & Olness, 2003](#)). All recruitment and consent procedures were approved by the Institutional Review Board (IRB) at Teachers College, Columbia University.

Data collection took place in a sound attenuated, electrically shielded room with the participant seated 80cm from a computer

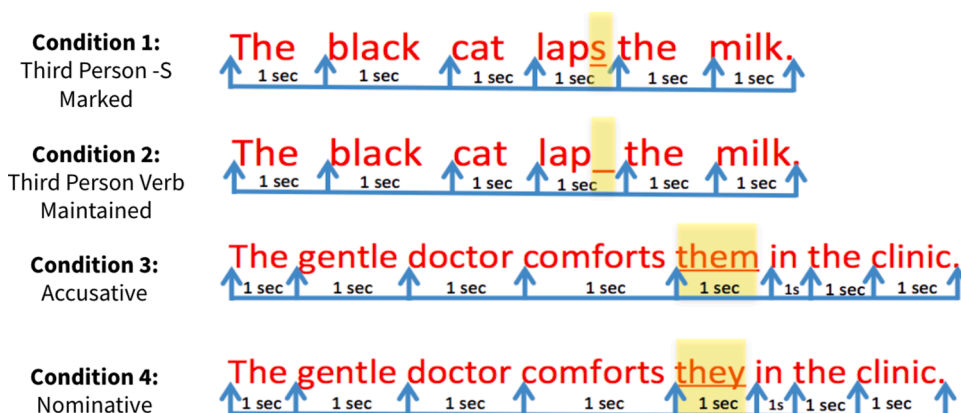


Fig. 1. Sample stimulus sentences. ERPs were segmented from EEG recordings that were time-locked to the end of the target word in *Block 1* (e.g., lap/_ or lap/s) and the beginning of the target word in *Block 2* (e.g., /them or /they).

screen. Auditory stimuli were played free field using a Tannoy OCV 6 full-bandwidth speaker centered 193 cm above the participant's chair. Stimuli were presented using Eprime v 2.0.8.90 (Psychology Software Tools Incorporated, Pittsburgh, Pennsylvania). At the start of each block, participants read on-screen instructions asking them to remain fixated on a crosshair while listening to sentences. At the end of each sentence, when the crosshair turned into a question mark, participants were asked to make an acceptability judgment via button press (Serial Response Tool, Psychology Software Tools Incorporated, Pittsburgh, PA).

2.2. Stimuli

The experimental stimuli consisted of two types of sentences with two constructions within each type, totaling 4 conditions. Each of the 2 main types were presented auditorily as two separate blocks, such that Conditions 1 and 2 (*Block 1*) were presented in a different grouping as Conditions 3 and 4 (*Block 2*; see [Table 1](#)). *Block 1* contained the 3rd Person Verb -s *Marked* and 3rd Person *Maintained Verb* conditions. *Block 2* contained the *Accusative Object* and the *Nominative Object* sentence constructions, respectively. Both of these constructions are non-contrastive for MAE and AAE speakers, since the *Accusative Object* condition is grammatical for both groups, and the *Nominative Object* condition is ungrammatical for both groups (see [Table 1](#) for block design with predicted outcomes).

2.2.1. Stimuli recording

All stimuli were created using natural speech recordings, from a 31-year-old, white Southern American, female voice-over professional. The recording artist is a bidialectal speaker of MAE and Southern White English (SWE), which has documented phonetic and grammatical overlaps with African American English ([Oetting, 2015](#)). The recording artist was instructed to speak in their natural manner and read a list of words by grammatical category to control for effects of intonation and inflection. Words were subsequently spliced together into sentences, with one-word-per-second presentation, resulting in 6 s trials for *Block 1* and 8 s trials for *Block 2* (see [Fig. 1](#)). The differences between word durations were spliced with intervals of silence using Praat v5.3.2 software ([Boersma & Wenink, 2005](#); [Wood, 2005](#)). Auditory stimuli were normalized to 70 dB SPL using Audacity software ([Mazzoni, 2008](#)) and verified to be 70 dB SPL on the stimulus presentation computer using a sound pressure level meter.

2.3. Experimental tasks

2.3.1. Block 1: contrastive stimuli

In *Block 1*, sentence-length stimuli consisted of six monosyllabic words, presented in the following sequence: 'The' + [adjective modifier] + [animate noun] + [present tense verb] + 'the' + [inanimate noun]. In Condition 1, the onset of the '-s' marker on target verbs was restricted to voiceless /s/, following a voiceless plosive consonant /p/, /k/ or /t/ (e.g., breaks, hits, laps). To control for possible differences across target items, stimuli in Condition 2 were the same recordings used in Condition 1, with the /s/ manually removed. Therefore, each trial served as its own control for lexical frequency and other psycholinguistic factors across *Block 1*. A total of 92 sentences were auditorily presented in *Block 1*, 23 per condition presented twice, in alternating blocks with *Block 2* (see full list of stimulus sentences in [Appendix A](#)).

2.3.2. Block 2: non-contrastive stimuli

In *Block 2*, stimuli consisted of 8 words presented in the following sequence: 'The' + [adjective modifier] + [animate noun] + [present tense verb] + [pronoun] + [preposition] + 'the' + [inanimate noun]. Again, to control for differences across target items, recordings in Conditions 3 and 4 were identical, differing only on pronoun type, which was manually inserted (see full list of stimulus sentences in [Appendix A](#)). In *Block 2*, a total of 132 sentences were auditorily presented, with 33 sentences per condition. *Block 2* was presented twice in alternating blocks with *Block 1*.

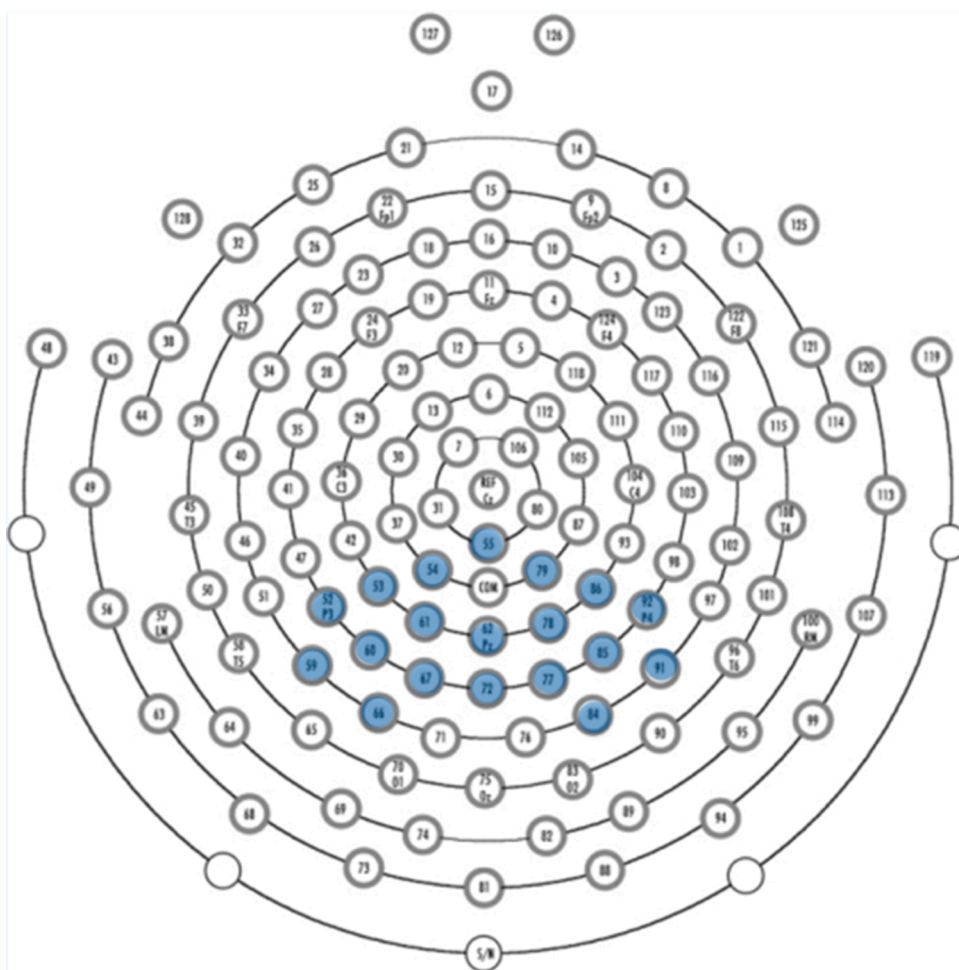


Fig. 2. Blue outlined electrodes were selected for the P600 montage analysis.

2.4. Electrophysiological recordings

Data in this experiment were collected via 128-channel geodesic sensor nets (Tucker, 1993; Magstim-Electrical Geodesics Inc.). Electrodes were adjusted and rehydrated with potassium chloride solution such that impedances did not exceed 40k Ω . Impedances were measured prior to the start of the experiment and re-measured between blocks. Magstim-Electrical Geodesics Incorporated (EGI) Netstation (v4.5.6) data acquisition software was used to record the continuous EEG with a sampling rate of 250 Hz. In order to prevent antialiasing of the signal, the raw EEG data were filtered using an analog 0.3–30 Hz low-pass filter based on the Nyquist frequency of the sampling rate, prior to digitization (filter settings for FIR Passband Gain: 99.0 % (-0.1 dB); Stopband Gain: 1.0 % (-40.0 dB); Rolloff: 2.00 Hz). Electrodes above and below the eyes, and at the outer canthi, allowed for identification of eye movement artifact.

2.5. Data analysis

2.5.1. Behavioral data

Behavioral data for response type and reaction time for the acceptability judgment task were averaged for each condition. The judgment task data were compared within and between groups, using two-tailed independent samples t-tests assuming unequal variance, to evaluate differences between response type and reaction time in each condition for each group of speakers.

2.5.1.1. Acceptability judgment task. The –s marked/maintained verb judgments in Block 1 were the target of interest in these conditions, where behavioral responses may reflect contrastive grammar between dialect groups or may reflect standard language bias by both monodialectal and bidialectal speakers. Behavioral responses in each condition were marked as 1 or 0 by Eprime software, such that average responses could be calculated as being either more correct/incorrect for non-contrastive sentences (-s marked), and more incorrect or MAE-biased for contrastive (maintained verb) sentences.

Responses to the case conditions in Block 2 were collected similarly, and represented a grammaticality judgment task where one

Table 2

Behavioral responses to judgment tasks in blocks 1 and 2

	Groups	'-s' Mark acceptability	Maintained verb acceptability	'-s' Mark RT	Maintained verb RT
Block 1	Monodialectal	0.963 (SD=.045)	0.906 * (SD=.147)	711.43ms (SD=354.28)	746.22ms (SD=382.35)
	Bidialectal	0.908 (SD=.136)	0.736 * (SD=.242)	771.33ms (SD=310.22)	712.44ms (SD=263.7)
		Accusative Object Acceptability	Nominative Object Acceptability	Accusative Object RT	Nominative Object RT
Block 2	Monodialectal	0.972 (SD=.046)	0.979* (SD=.029)	603.15ms (SD=319.36)	526.71ms** (SD=181.17)
	Bidialectal	0.905 (SD=.125)	0.913* (SD=.09)	763.26ms (SD=202.85)	726.91ms** (SD=206.43)

Note. For response type, the closer the score is to 1.0, the more “correct” or MAE-aligned they were. *Response types in the maintained verb condition were significantly different between groups ($t(26) = 2.289$, $p = .039$). **Acceptability judgments and reaction times (RT) for the nominative object (non-contrastive error) condition were significantly different between groups; ($t(25) = 2.537$, $p = .023$) and ($t(25) = -2.67$, $p = .013$), respectively.

condition contained a shared morphosyntactic violation that applies to both groups' language varieties.

2.5.2. EEG data pre-processing

A standard ERP analysis protocol was used to process the EEG data (Handy, 2005; Luck, 2005; Picton et al., 2000). Raw EEG recordings were digitally re-filtered offline using a 30 Hz LowPass filter and .3Hz HighPass filter (FIR Passband Gain: 99.0 % (-0.1 dB); Stopband Gain: 1.0 % (-40.0 dB); Rolloff: 2.00 Hz), and then subjected to automatic and manual artifact rejection protocols. Eye channels and noisy channels were marked as *bad*, and interpolated using spherical spline interpolation based on recorded data from surrounding sensors. Data were re-referenced offline to an average reference (Handy, 2005).

2.5.3. Extraction of ERP components

The continuous recordings were segmented into 1200 ms epochs, including 100 ms pre-stimulus onset and 1100 ms following the onset of target stimuli. A time window of 500-800 ms post stimulus onset was targeted for identification of the P600 ERP component. Segments from each condition were averaged together to enhance the signal-to-noise ratio, and to identify time-locked event-related responses associated with onset of the target. EEG epochs were averaged separately for each of the four conditions, for each individual participant. The average waveforms were baseline-corrected, by subtracting the average electrical potential during the 100ms baseline period from the epoch of interest.

Analyses were constrained to a montage of centroparietal electrodes (see Fig. 2). Auditory P600 signals have been shown to have multiple generators and are maximally recorded at centroparietal electrodes (Handy, 2005; Luck, 2005). Channels for the P600 montage were selected based on prior research investigating this component in AE dialects and included electrodes 52, 53, 54, 59, 60, 61, 66, 67, 72, 77, 78, 79, 80, 85, 86, 87, 92, 93 in the 128-channel geodesic sensor net by EGI (Conrey et al., 2005).

2.5.4. EEG data analysis

After pre-processing in NetStation, EEG data were exported for statistical analyses in the R Studio (v3.2.2) data analysis package. Repeated measures analyses of variance (ANOVAs) were conducted to test for main effects and interactions in the data (cf. Dien & Santuzzi, 2005). A 2×2 balanced factorial design was used with factors Spoken Dialect (Monodialectal and Bidialectal groups), and 4 levels of Sentence Condition which were subdivided into 2 blocks. Data were analyzed between conditions in Block 1 (Sentence Types 1 and 2) and Block 2 (Sentence Types 3 and 4), as well as between groups (Monodialectal and Bidialectal). The ANOVAs were followed by planned comparisons (paired samples and independent samples t-tests) at each level of each significant variable.

Behavioral data were analyzed for proportions of correct responses in Condition 1, MAE-biased responses in Condition 2, and correct responses in Conditions 3 and 4 across both groups. Differences were evaluated using a repeated measures ANOVA with a within groups factor of Sentence Type and a between groups factor of Dialect for both Block 1 and Block 2. Response times (RTs) were analyzed using t-tests, after timed-out responses (i.e., *no response* trials) were removed.

3. Results

3.1. Behavioral results

Behavioral data for response type and reaction time were collected from 13 AAE-MAE bidialectal speakers and 15 MAE monodialectal speakers. Two participants were omitted from behavioral analyses for *non-responses* to sentences exceeding 50% of trials during the judgment task. Since evidence from the behavioral analysis is not being correlated to neurophysiological responses, removal from behavioral analysis did not necessitate removal from ERP analysis for these participants.

3.1.1. Block 1 response types and reaction times for judgment task

An independent samples t-test was applied to the scores in each group for each condition in Block 1. In condition 1 no significant difference in acceptability judgment was expected between groups and this prediction was supported ($t(26) = 1.4$, $p = .183$). In condition 2 it was again predicted that behavioral responses would not differ, based on the possibility that bidialectal listeners provide MAE-aligned responses. However, this prediction was contradicted by the findings, which showed a significant difference ($t(26) = 2.289$, $p = .039$). This indicates that bidialectal speakers selected ‘acceptable’ more often than monodialectal speakers for sentences

Table 3Mean amplitudes (μV) in the 400–800ms time window.

	Group	Mean Amplitude / SD -s mark	Mean Amplitude / SD Maintained Verb
Block 1	Bidialectal (n = 14)	0.567 μV , SD = 1.258	0.748 μV *, SD = 0.826
	Monodialectal (n = 13)	0.779 μV , SD = 1.132	1.847 μV *, SD = 1.401
		Mean Amplitude / SD Accusative Object	Mean Amplitude / SD Nominative Object
Block 2	Bidialectal (n = 14)	-2.80 μV , SD = 0.783	0.486 μV , SD = 0.667
	Monodialectal (n = 13)	-0.398 μV , SD = 1.121	0.469 μV , SD = 1.029

Note. *Only mean amplitudes in the maintained verb condition were significantly different between groups ($t(25) = 2.503, p = .019$).

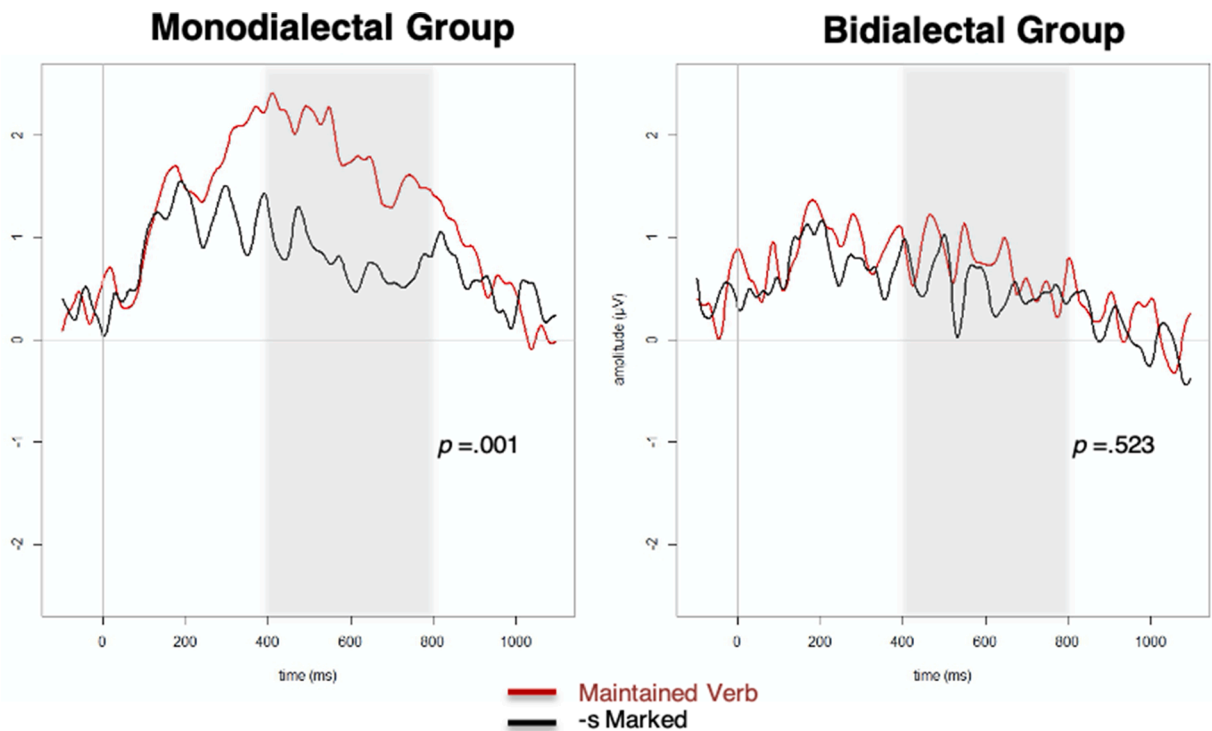


Fig. 3. Grand Average Waveform in [Block 1](#). Monodialectal and bidialectal responses are shown from 0–1100ms post-target stimulus onset. Repeated measures ANOVA between conditions indicated a significant P600 effect for the monodialectal group ($p = .001$).

where the verb was maintained in the present progressive tense. Average response times were not significantly different between groups in either condition (see [Table 2](#) below).

3.1.2. Block 2 response types and reaction times for acceptability judgment task

An independent samples t-test was applied to evaluate differences between the groups for each condition in [Block 2](#). In conditions 3 (accusative object) and 4 (nominative object), the sentence types were predicted to elicit similar results between groups since both are non-contrastive between dialects. This prediction was contradicted, and significant differences of response type were found between groups in the *nominative object* condition ($t(25) = 2.537, p = .023$), but not the *accusative object* condition, though the findings may be leaning toward significance ($t(25) = 1.831, p = .087$). This finding shows that bidialectal speakers were more likely to rate nominative object sentence types as “acceptable” compared to monodialectal speakers. A significant difference in response time between groups was also found in the *nominative object* condition ($t(25) = 2.537, p = .023$) but not the *accusative object* condition ($t(25) = -1.566, p = .131$), suggesting that bidialectal speakers took longer to make decisions about sentences with a nominative object, relative to monodialectal speakers (see average acceptability and response times in [Table 2](#)).

3.2. Neurophysiological results

ERP results are based on data collected from 14 bidialectal (AAE-MAE) and 13 monodialectal (MAE) participants. Analyses of the P600 component are reported between groups (monodialectal/bidialectal) for each block, with alpha level set to 0.05. Split-plot repeated-measures ANOVAs on mean amplitudes were conducted to analyze effects and interactions for factors SENTENCE TYPE

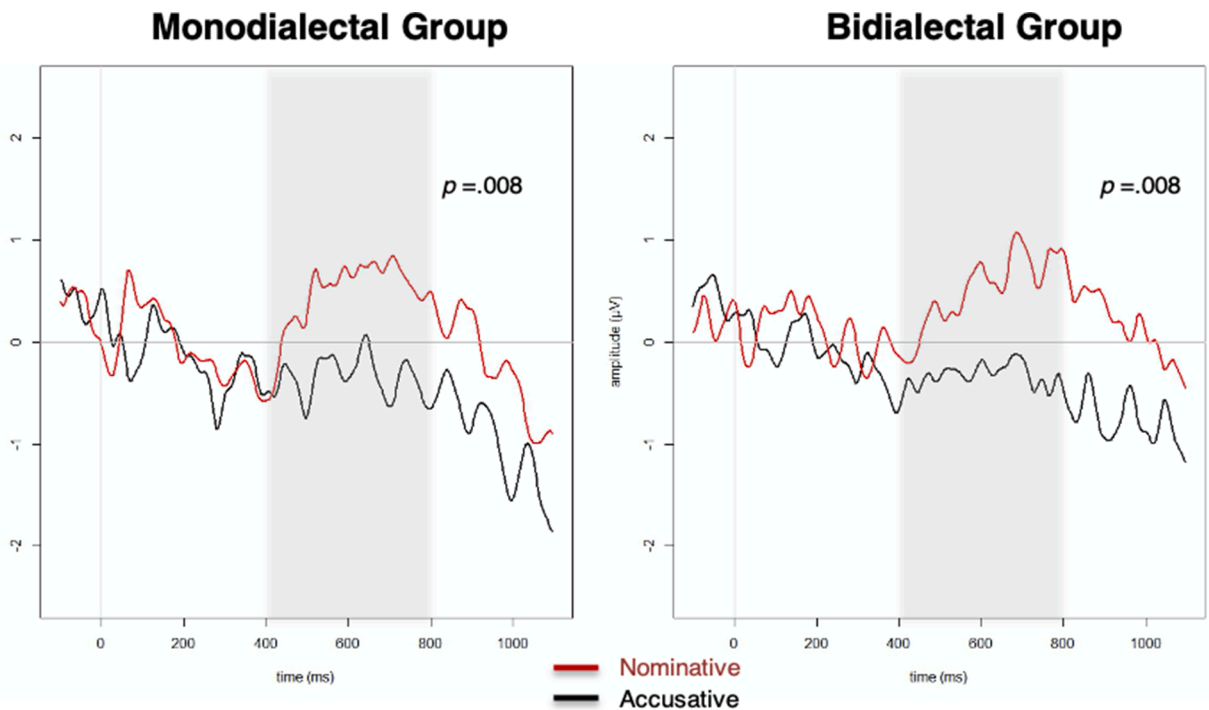


Fig. 4. Grand averaged waveforms for Block 2. Both groups show a similar P600 response to the nominative object condition.

(-s marked/maintained verb and nominative/accusative objects) and GROUP (bidialectal and monodialectal).

3.2.1. Block 1 P600

A two-way analysis of variance having one between-subjects factor (GROUP) with two levels (monodialectal/bidialectal) and one within-subjects factor (SENTENCE TYPE) with 2 levels (-s marked/maintained verb) was applied to the mean amplitudes per condition for each participant within each group. The main effect of *sentence type* was significant ($F(1, 25) = 10.837, p = .003$), indicating a difference in mean amplitudes between conditions. The interaction between *group* and *sentence type* was also significant ($F(1, 25) = 5.461, p = .028$). Follow-up independent samples t-tests evaluating mean amplitude differences for each condition between groups revealed a significant difference between group means in the maintained verb condition ($t(25) = 2.503, p = .019$), pointing to a larger P600 amplitude in response to maintained verb forms from the monodialectal group. No significant differences were noted between groups in the -s marked condition ($t(25) = .456, p = .652$), indicating that this shared morphosyntactic rule was processed similarly for all participants. Table 3 provides descriptive statistics and Fig. 3 shows grand average waveforms plotted for each group in each condition.

3.2.2. Block 2 P600

A two-way ANOVA having one between-subjects factor (GROUP) with two levels (monodialectal/ bidialectal) and 1 within-subjects factor (SENTENCE TYPE) with 2 levels (accusative/nominative objects) was again performed for the Block 2 data. The main effect of *sentence type* was significant ($F(1, 25) = 19.830, p < .001$). The interaction between *group* and *sentence type* was not significant ($F(1, 25) = .076, p = .785$). Follow-up independent samples t-tests comparing mean amplitudes in each condition revealed no significant differences between group means for either sentence type (accusative object: $t(25) = -.320, p = .751$; nominative object: $t(25) = -.053, p = .959$). Table 3 provides descriptive statistics and Fig. 4 shows grand average waveforms for each group in response to the accusative and nominative object conditions.

4. Discussion

This study investigated how AAE-MAE bidialectal and MAE monodialectal speakers respond to morphosyntactically contrastive and non-contrastive sentence types, in order to gain insight into how individuals process grammatical differences between dialects online, as well as during decision-making tasks.

4.1. Interpretation of ERP findings

It was predicted that bidialectal speakers would not demonstrate a P600 response to sentences containing an AAE morphosyntax

Block 1

Block 1 Sentences

The	fat	aunt	sweep/s	the	room.
The	black	cat	lap/s	the	milk.
The	pink	pig	take/s	the	food.
The	sly	dog	track/s	the	scent.
The	old	man	fight/s	the	boy.
The	slow	friend	pack/s	the	bag.
The	fast	car	hit/s	the	truck.
The	slow	chef	cook/s	the	pie.
The	calm	mom	knit/s	the	sock.
The	rich	cook	chop/s	the	pear.
The	sick	cow	eat/s	the	grass.
The	thin	monk	write/s	the	book.
The	big	maid	tap/s	the	desk.
The	mad	wife	break/s	the	cup.
The	short	teen	kick/s	the	ball.
The	cute	kid	pop/s	the	gum.
The	shy	niece	make/s	the	speech.
The	smart	clerk	wrap/s	the	box.
The	wet	mop	wipe/s	the	floor.
The	sweet	girl	pick/s	the	rose.
The	strong	dad	rake/s	the	leaves.
The	young	child	light/s	the	fire.
The	small	son	lick/s	the	spoon.

Block 2

Block 2 Sentences

The	anxious	bridesmaid	hurries	them/they	to	the	altar.
The	clever	student	prepares	her/she	for	the	exam.
The	pretty	lady	awaits	him/he	at	the	lobby
The	clever	student	prepares	her/she	for	the	exam.
The	healthy	jogger	outruns	them/they	at	the	racetrack.
The	angry	boatman	commands	him/he	to	the	engine.
The	bankrupt	chairman	receives	us/we	in	the	office.
The	sneaky	actor	admits	her/she	from	the	sidedoor
The	massive	awning	protects	us/we	from	the	downpour.
The	cheery	sophomore	welcomes	him/he	to	the	clubhouse.
The	greedy	woman	marries	him/he	for	the	money.
The	bossy	grandchild	orders	her/she	to	the	playground.
The	gentle	doctor	comforts	them/they	in	the	clinic.
The	busy	lawyer	escapes	them/they	in	the	bathroom.
The	eager	puppy	follows	us/we	to	the	kitchen.
The	cranky	baby	distracts	her/she	from	the	party.
The	bloody	patient	infects	them/they	with	the	disease.
The	awkward	husband	questions	him/he	at	the	driveway.
The	brunette	diva	annoys	us/we	in	the	movie.
The	muscly	climber	carries	her/she	down	the	mountain.
The	distant	fire	reaches	them/they	in	the	morning.
The	witty	mayor	persuades	him/he	at	the	debate.
The	foreign	buyer	outbids	her/she	at	the	auction.
The	wild	artist	blindfolds	them/they	at	the	concert.
The	bitter	actress	applauds	him/he	for	the	award.
The	absent	grandson	avoids	them/they	in	the	attic.
The	crafty	gambler	swindles	him/he	at	the	tavern.
The	agile	gymnast	baffles	them/they	at	the	circus.
The	handsome	linguist	bothers	her/she	at	the	lecture.
The	able	midwife	nurtures	her/she	at	the	birthing.
The	deadly	fighter	attacks	him/he	in	the	alley.
The	friendly	beagle	cuddles	us/we	on	the	sofa.
The	quiet	inmate	tattoos	him/he	in	the	prison.

feature, based on the hypothesis that dialectal language varieties are distinctly processed, and this was supported by the findings. Neither group showed a significant difference in amplitude when listening to non-contrasting (–s marked) sentences, but a P600 signature was found in the MAE-only group when listening to maintained verb sentences ($p = .019$). This finding indicates that a reanalysis/repair of maintained verb sentences is being called upon by the monodialectal group alone, since it is a known feature of AAE in the bidialectal group. This online detection of anomaly by the monodialectal group aligns with findings reported by [Conrey et al., \(2005\)](#), whose unmerged dialect speakers produced a P600 response to contrasting auditory and text-based presentations of

“pin/pen” type items, suggesting that such differences are processed differently by monodialectal and bidialectal speakers. In studies with sentence-length presentations of NMAE dialects however, the online detection process was seen to be the same across conditions, including contrasting ones (Weissler & Brennan, 2020; Zaharchuk et al., 2021). The present investigation may have allowed for a different finding by selecting a highly prevalent feature of a dialect, which might make its usage common enough to be processed differently (perhaps more automatically) than less frequently occurring forms, like the auxiliary “be” contrasts (e.g., He got sick, so he coughing a lot) or the double modal (e.g., might could) targeted by previous investigations. In the current study then, standard features may dominate online language processing, but increased usage might have altered expectations and reduced the need for reanalysis or repair, as reflected in the ERP responses from the bidialectal speakers in the current investigation (Sumner et al., 2014). One limitation of this investigation, therefore, may be that different dialect-specific constructions will show varying response types from bidialectal speakers that align more or less with monodialectal MAE speakers’ neurophysiological responses. Future research using similarly controlled groups of language users but presenting different contrasting features of high and low frequency would shed light on this possibility.

A set of non-contrasting constructions were also presented as control conditions, predicted to elicit ERP responses that would not differ significantly between groups based on the hypothesis that language varieties with overlapping features are processed similarly between speakers (Khamis-Dakwar & Froud, 2007; Zaharchuk et al., 2021). Since a P600 was elicited for both groups in response to the non-contrastive sentence types, this position was supported, strengthening evidence that sentence processing loads are similar for overlapping grammatical constructions across dialects.

4.2. Interpretation of behavioral responses

With regard to behavioral responses, it was predicted that there would be no significant difference between groups’ acceptability responses, but that reaction times would be greater in bidialectal speakers for the maintained verb sentence type. The acceptability prediction was based on the hypothesis that there is a dissociation between automatic language processing and more conscious decisions about grammaticality, which led to the prediction that bidialectal speakers would accept only MAE-aligned sentences because of the “high prestige” language setting of the university lab (Adger et al., 2014). Unexpectedly, bidialectal speakers showed significantly greater acceptance of maintained verb sentences ($p = .039$), and the nominative object sentences ($p = .023$), the latter of which is not a contrasting feature of AAE and MAE. The increased latency prediction was based on the hypothesis that increased cognitive load during decision-making tasks is reflected by increased response time, and again, the opposite was seen here, with no difference across conditions between groups.

Increased acceptance of dialectal features in a grammaticality judgment task has been similarly reported in similar investigations with low prestige dialect varieties, however, suggesting that such judgments are not so easily biased by environments known to use a standard language (Conrey et al., 2005; Khamis-Dakwar & Froud, 2007; Zaharchuk et al., 2021). Interestingly, in the control conditions of Block 2, bidialectal listeners showed increased reaction times overall (i.e., to both sentence types), in addition to the increased acceptability responses to a shared error (i.e., *nominative object* sentence types). This finding is being tentatively interpreted as a limitation related to greater cognitive demands that could lead to hyper- or hypo- correction for grammatical decision-making situations where there is the *possibility* of a contrast between dialects. This possibility is thought to exist here because while the presented shared error condition did not contain an AAE feature (e.g., The gentle doctor comforts he in the clinic), a similar construction of pronoun case in the subject position (e.g., Him going upstairs) is a feature for some bidialectal speakers of AAE. Dialect-specific pronoun features in AAE have been assigned various descriptions in the literature, including *appositive pronouns* (e.g., “That teacher, she yell at the kids”); *benefactive datives* (e.g., “I got me a new car”); *demonstrative ‘them’* (e.g., “I love them shoes”); and *‘undifferentiated pronoun case’* (e.g., “I don’t like them bad people”) (e.g., Rickford, 1999; Washington & Craig, 2002; Wolfram, 2004).

The major implication here – which seems of particular interest to clinicians and educators – is that while adult bidialectal speakers may accept AAE sentences with highly familiar features just as quickly as MAE sentences, grammaticality judgments of MAE sentences that contain non-contrastive features *similar to* AAE-specific features are less accurate and take more time. This would indicate that despite automatic detection of syntactic errors, a conscious or unconscious recognition of the similarity to other dialectal features can lead to differences between monodialectal and bidialectal speakers. This evidence offers important considerations for bidialectal speakers who may be misjudged as having difficulties with non-contrastive MAE features when these are presented in sentences containing non-contrastive errors that are *almost* like sentences with contrastive AAE features. Anecdotal evidence for this was also apparent in the responses of bidialectal participants to a debrief interview, where several reported “losing track” of whether what they had just heard was acceptable or not. Given this, the need for assessments that offer more than pass/fail trials for language with and without dialect features is clear (Oetting et al., 2019).

5. Conclusions

We present evidence that, when the conceptualized distance between two language varieties is minimal or nonexistent, differences in neurophysiological and behavioral responses can still be seen. The types of different responses between bidialectal and monodialectal speakers may, however, rely on the frequency of the given features (contrastive or non-contrastive) and their similarity to other, familiar dialectal-specific constructions. Research with NMAE varieties is in the earliest of stage of investigation, and many unresolved issues remain. Our findings point to the absolute need for more research on how automatic language processing, as well as language performance, in dual-language users may differ where varieties are sociolinguistically conceived as being part of a single language. In combination with previous research findings, this study suggests that choosing participants based on qualifications for

standard language variety use alone may pose a risk to neurophysiological and behavioral investigations of morphosyntax, even when target stimuli contain only non-contrasting/dialect-general features. Finally, future research must pose the same questions and the same level of rigor for investigations of bidialectal language performance as are directed at understanding so-called standard language bilingualism, in order to better understand the nature of concepts like codeswitching and error within and between these language varieties.

CRediT authorship contribution statement

Felicidad M. Garcia: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing. **Guannan Shen:** Methodology, Visualization, Writing – original draft, Writing – review & editing. **Trey Avery:** Methodology, Validation, Writing – review & editing. **Heather L. Green:** Methodology, Writing – review & editing. **Paula Godoy:** Project administration, Writing – review & editing. **Reem Khamis:** Supervision, Methodology, Funding acquisition. **Karen Froud:** Supervision, Methodology, Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

Acknowledgments

This work was supported by many mentors, colleagues, and financial support from generous sources, including the Teachers College (TC) Doctoral Dissertation Grant, the Scholl Scholarship Award, the TC Vice President's Student Research in Diversity Grant, and the ASHA Advancing Academic Research Careers (AARC) award. The authors thankfully acknowledge the work of Black scholars, activists and artists, who led various language-related and education-based fields to recognize, respect and celebrate Black English, paving the way for this study to take place.

Appendix A: Sentence stimuli

Block 1

Block 2

Appendix B: Demographic & language background questionnaire

Date:

Age:

Gender: Male / Female / Trans / Other:

Race:

Ethnicity:

I consider my childhood socioeconomic income status as most closely resembling*:

- a Poverty
- b Working Class
- c Lower Middle
- d Middle
- e Upper Middle
- f Upper
- g I don't know / I would rather not say.

As a child, I was raised in a family with an average total income most closely resembling the following bracket*:

- a \$23,050 or less
- b \$23,050 - \$32,500
- c \$32,500 - \$60,000
- d \$60,000 - \$100,000
- e \$100,000 - \$150,000
- f <\$150,000
- g I don't know / I would rather not say

I consider my current socioeconomic income status as most closely resembling*:

- a Poverty

- b Working Class
- c Lower Middle
- d Middle
- e Upper Middle
- f Upper
- g I don't know / I would rather not say.

As an adult, my income bracket most closely resembles the following level*:

- a \$23,050 or less
- b \$23,050 - \$32,500
- c \$32,500 - \$60,000
- d \$60,000 - \$100,000
- e \$100,000 - \$150,000
- f <\$150,000)
- g I don't know / I would rather not say

*Income classes defined by the U.S. Census 2012, as described in this article: <http://bit.ly/23SQnnP>

What is your highest level of education started? (Please also list any diplomas, degrees, vocational license, certificates, etc):

What is your current occupation?

Are you right or left handed? (Please report your laterality index and right/left decile from online questionnaire)

Any history of any of the following? (Write 'yes' if so.)

- Speech or language disorder?
- Reading impairment/disability?
 - Attention deficit disorder?
 - Neurological condition (e.g., epilepsy, schizophrenia, depression)?
 - Vision deficit?
- Deaf or Hard of Hearing?
- Use of medication? Please indicate type:
- Other: __

What languages do you speak?

Which languages have you been exposed to?

What city (cities) did you live in from the ages of 0-15?

What neighborhood(s)?

Debriefing Questions

In the United States, individual speakers of English may speak different regional dialects depending on geographic location and cultural background. Language research has embraced the study of this rich diversity only in recent years, and has recognized the importance of all English dialects as equal linguistic systems. These conclusions have influenced our nation's educational system for the better. We hope our study will continue to contribute to this research.

Current research in language and literacy has been expanded to include non-mainstream American English (NMAE) dialects other than AAE such as Southern American English, Creole English, Appalachian English, and Latino English.

The following questions are meant to get a clearer picture of each participant's unique language background.

That you know of, what English dialects do you speak or have you been exposed to? (Mark all that apply)

- African American English
- Standard American English
- Southern English
- Chicano-American English (Mexican English)
- Canadian English
- Spanish-Influenced English
- Appalachian English
- Other non-mainstream English: _____

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